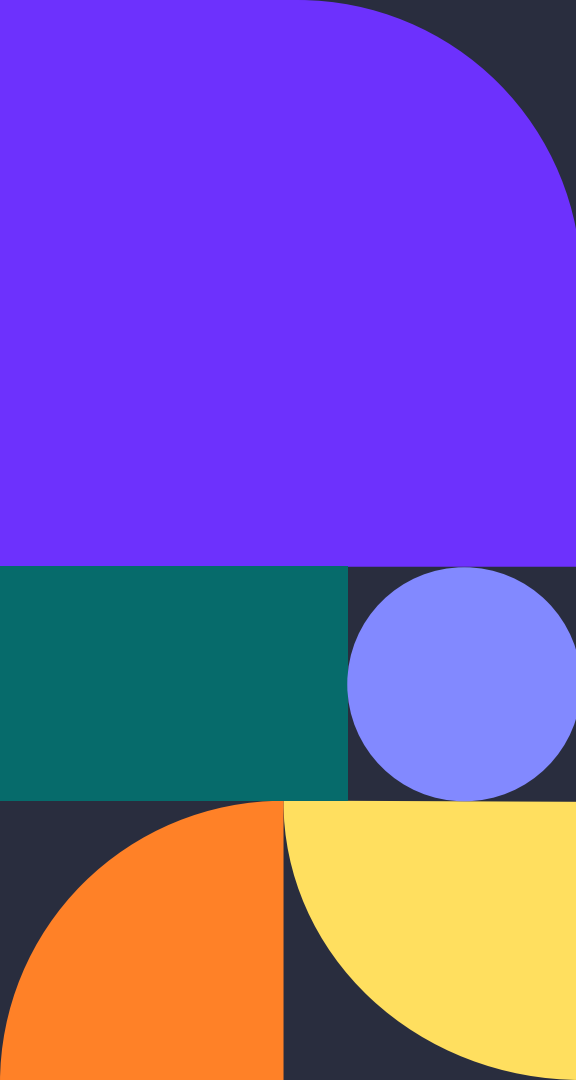




Finding Titan – A Proton Precession Magnetometer for Underwater Magnetic Surveying

Michael Krueger, Mason Lebens, Ryan Nowak, Prakul Swaminath

Background

The University of Wisconsin participates in the Missing in Action Recovery and Identification Project (UW MIA RIP). Which works to recover, identify, and repatriate MIA U.S. service members by conducting annual field excavations and year-round research assistance to families of missing service members.

The focus of the proton magnetometer towfish is to locate downed aircraft and ships. Doing so aids in the identification and recovery of lost servicemembers in the shallow waters of the Vietnam coast.

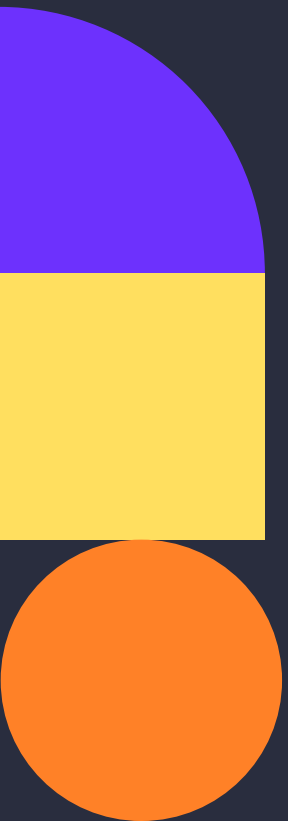
Project Goals



01. Build a
Magnetometer with
<1nT of sensitivity
03. Integrate
magnetometer with
GPS and build a
towfish
05. Survey the
Vietnam ocean floor
to recover MIA
soldiers

02. Design and build
custom PCBs

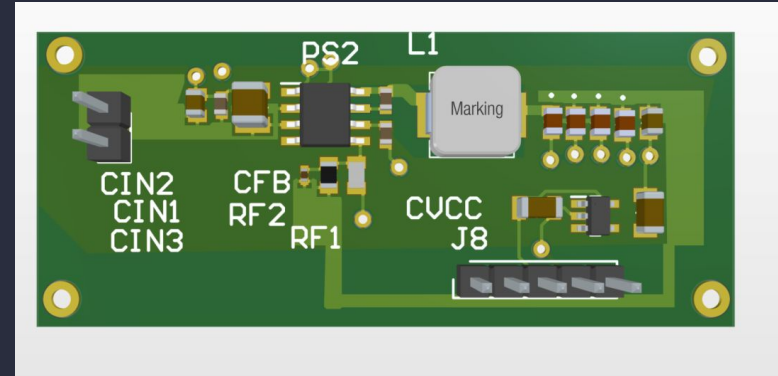
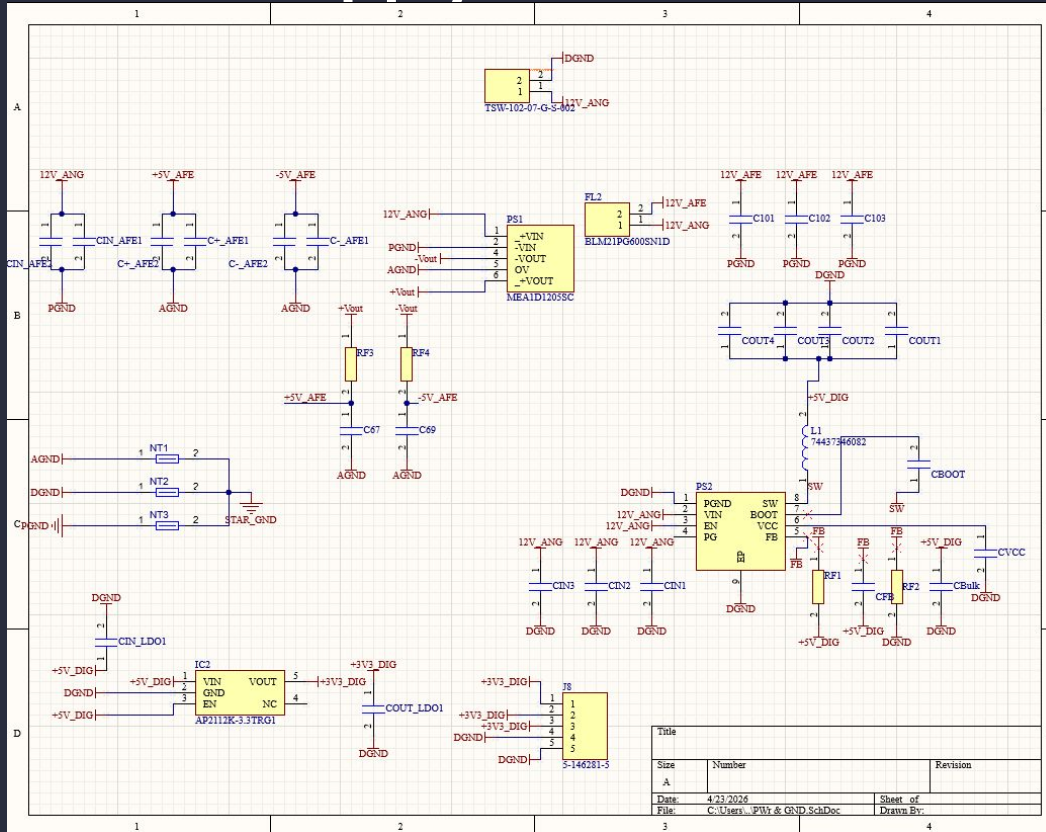
04. Map the
magnetic field over
frozen lakes



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01. **Circuit Design and Calculations**
02. **Algorithms**
03. **CAD Models and Towfish Design**
04. **Results**
05. **Challenges Faced**
06. **Future Considerations**
07. **Bill of Materials**

Power Supply and Ground Rails



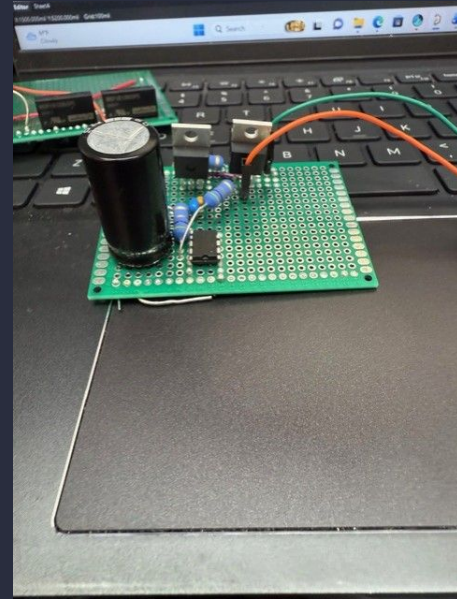
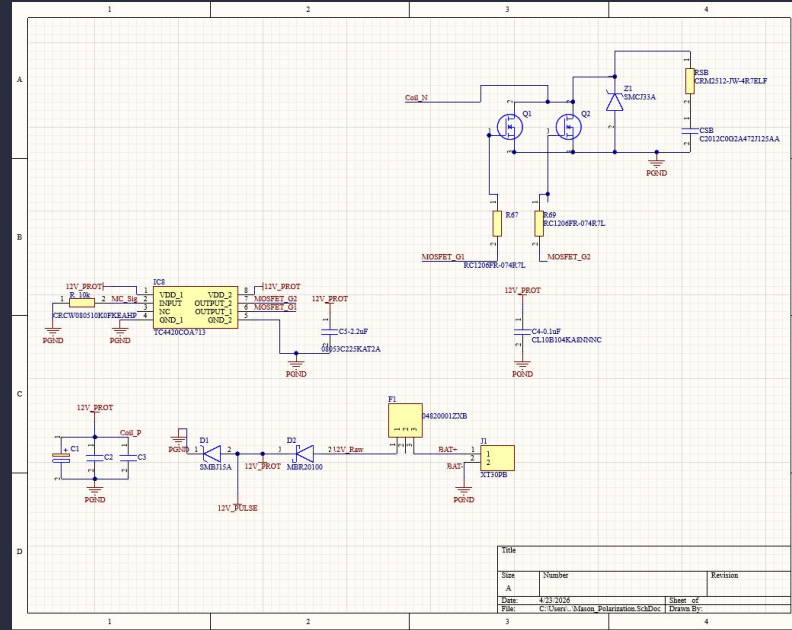
The magnetometer uses separated power rails to reduce noise between the pulse, digital, and analog circuitry. The 12 V input supplies the high-current polarization pulse, while the 5 V and 3.3 V rails power the MCU and digital circuitry. The analog components are powered from ± 5 V rails.

Separate PGND, DGND, and AGND regions keep pulse currents away from the sensing circuitry. These grounds connect only at a star-ground point.

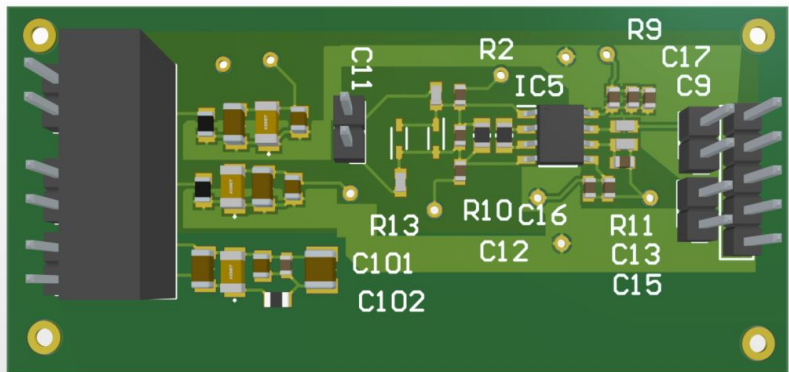
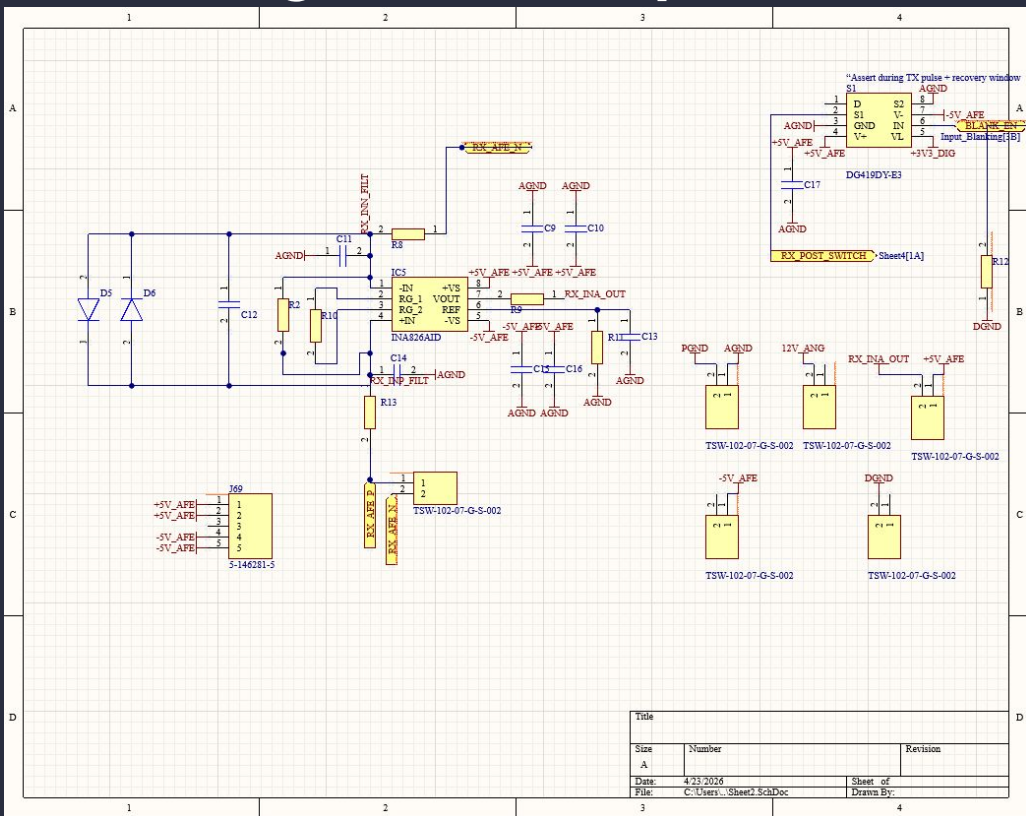
Coil Polarization

The polarization circuit takes a 3.3V input signal from the MCU to turn on and off a gate driver which allows current to flow through 2 redundant MOSFETs connected in parallel. These MOSFETs act as switches to turn on and off the 12V and 10.9A of current flowing through the polarization coil. The coil needs to have the least amount of ringing possible to produce the most accurate measurement. That is why connected across the drain and source of the MOSFETs is a TVS clamping diode and RC snubber.

Prior to the polarization coil there are 2 decoupling capacitors, a bulk capacitor to provide the large current pulse and a small high frequency bypass capacitor.



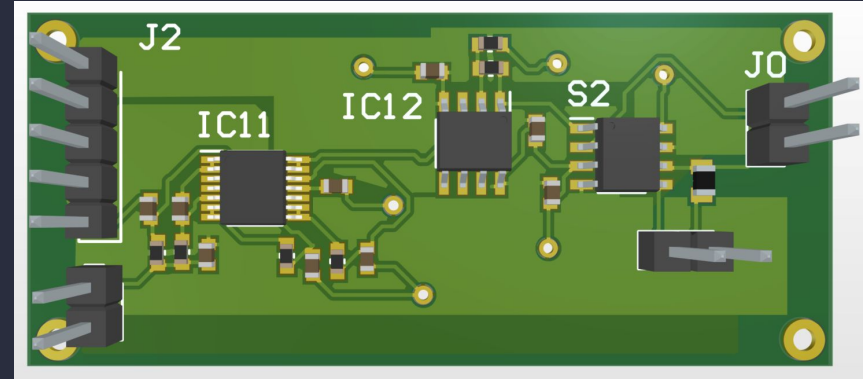
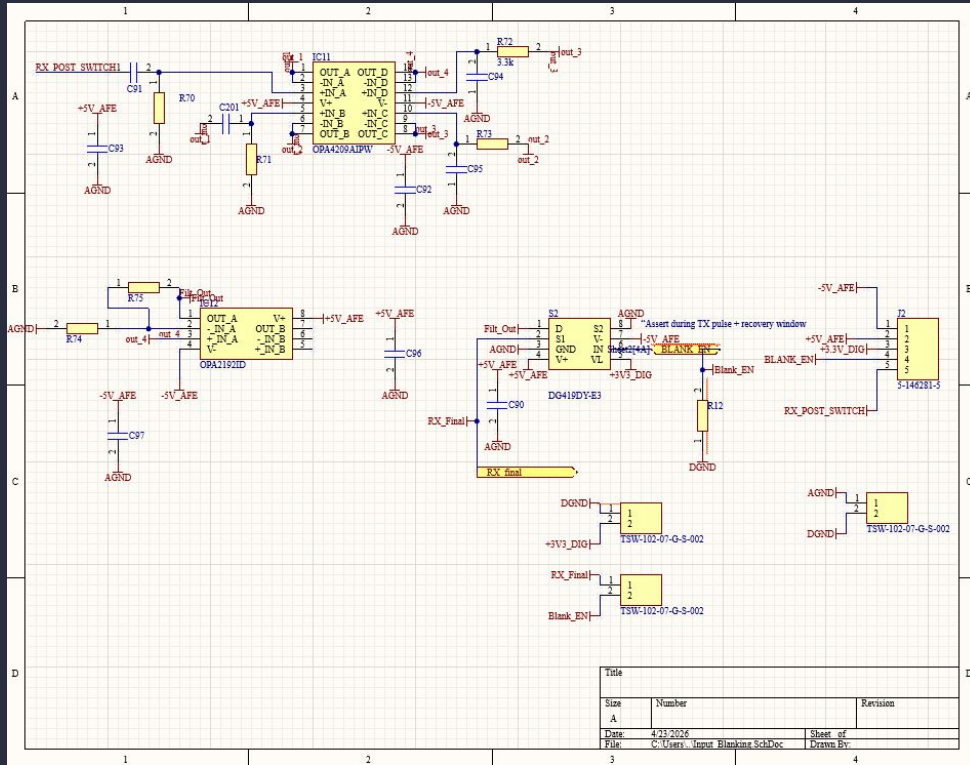
Blanking & Pre-Amplification



This stage protects the analog circuitry during the high-current polarization pulse. A DG419 analog switch blanks the receiver path during pulse events to prevent ringing and large transients from saturating the amplifier.

After blanking, the sensing coil signal is sent into a low-noise preamplifier stage. The pre-amp provides initial gain before filtering and ADC conversion, helping protect the small proton signal while limiting noise pickup.

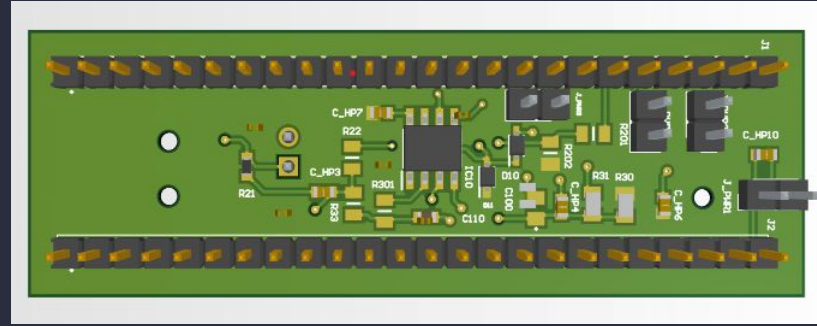
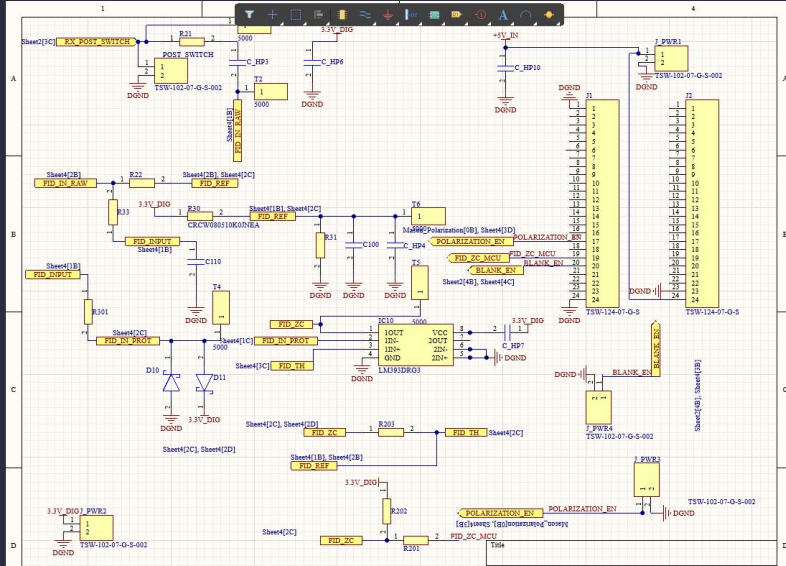
Filtering & Post Amplifier



This stage further conditions the signal after the pre-amplifier. Active filtering removes unwanted noise outside of the 800 to 4kHz pass band.

The post-amplifier then provides additional gain so the proton signal is large enough for ADC measurement. This stage provides 100X gain for a total gain of 2000x throughout the analog chain.

Digital Circuitry Design – How this works



This stage takes in the small, filtered, noisy AC proton precession signal and converts it into a digital wave (square wave) for the MCU to measure the frequency.

The MCU used is the Teensy 4.1 that was used from last semester. It calculates the time at the zero crossing edges and derives our frequency which is our Larmor Frequency.

Deriving the frequency, the MCU is able to calculate the magnetic sensitivity by taking the measured frequency and dividing it by the gyromagnetic ratio of protons (42.58Mhz/F).

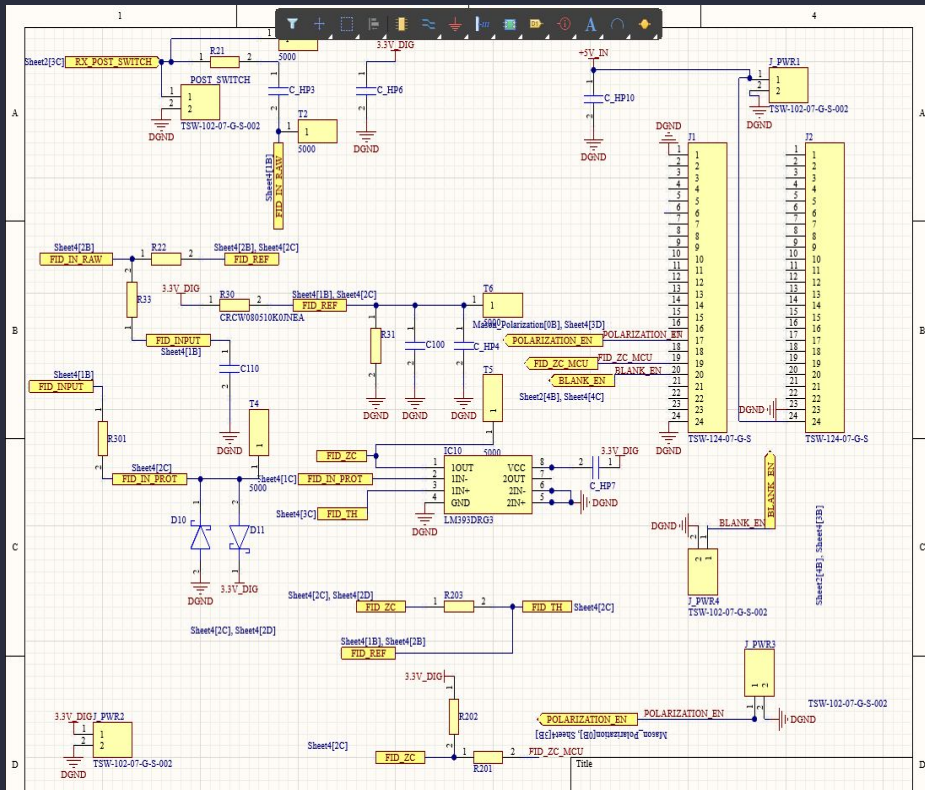
$B \text{ (nT)} = \text{frequency measured by MCU} / 42.58\text{Mhz/F}$

Note:

J1 & J2 are pins for the Teensy 4.1 MCU to be soldered on.

The pins labelled will be explained on the next slide.

J1 & J2 pin Layout Explained



Pin labelled **FID_ZC_MCU (pin 19)** is the filtered, proton precession signal from the AC sine wave into a square wave that goes into the MCU.

Pin labelled **POLARIZATION_EN (pin 17)** is the signal that the MCU sends the pulse controller board (look at previous slide on this) to drive the MOSFETS on/off.

Pin labelled **BLANKING_EN (pin 20)** is the blanking signal and should be before the analog circuitry right after the coils to prevent LC ringing, and "ignore the initial noise of the produced signal. Goal is to only pass the small FID decaying sinusoidal signal to the analog circuitry.

Pin labelled **JWR1** is the +5V power supply. This is electrically connected to **pin 24 on J2**. To provide +5V, need to connect a wire from the +5V rail from the power board into this pin.

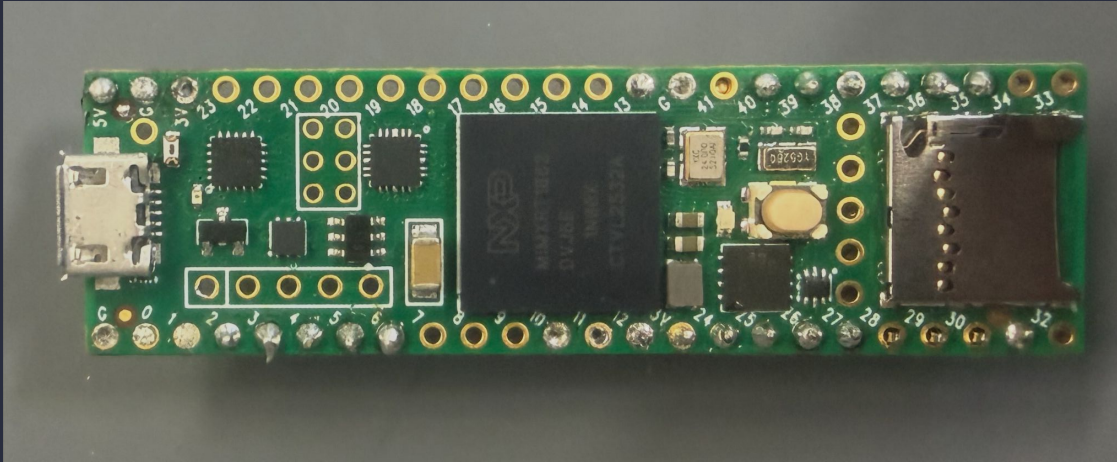
Pin labelled **JWR3** is the Polarization_EN, electrically connected to **pin 17 on J1**. Connect a wire from this pin to the correct pin on the MOSFET on the pulse controller board to provide blanking.

Pin labelled **JWR4** is the BLANKING_EN, electrically connected to **pin 20 on J1**. Connect a wire from this pin to right after the coils before⁰ the analog circuitry.

Note: Refer previous slides for blanking & polarization connections

Prakul Swaminath

Pin Layout on the MCU Teensy 4.1



How to solder MCU?

Solder the G pin on the MCU (top left on Arduino side) to J1 pin 24. That should give you all the correct connections.

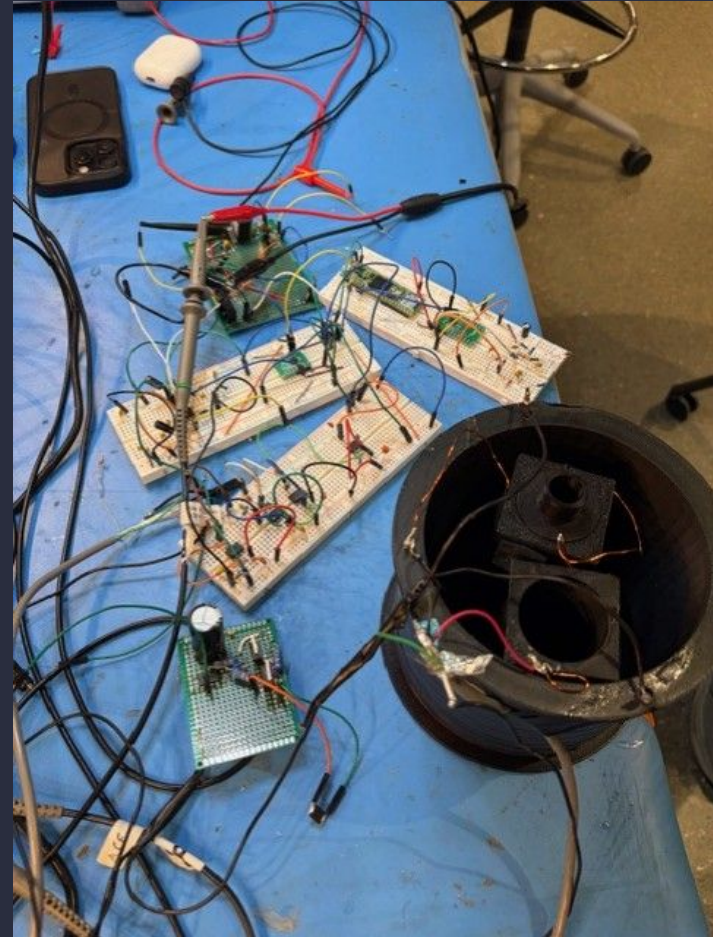
Pin 4 on Teensy 4.1 MCU is the: FID_ZC_MCU (square wave input signal) - This is pin 19 on J1 (schematic).

Pin 3 on Teensy 4.1 MCU is the BLANKING_EN (reduce LC ringing) - This is pin 20 on J1 (schematic).

Pin 6 on Teensy 4.1 MCU is the POLARIZATION_EN (drives MOSFETs) - This is pin 17 on J1 (schematic)

Assembled Circuit

Shown to the right is the power supply/ground rails, coil polarization, blanking/pre amplification, filtering/post amplification, and MCU circuit boards. Also shown is the 5.5" polarization coil and both 2" sensing coils.



Coil Design and Calculations

Polarizing Coil:

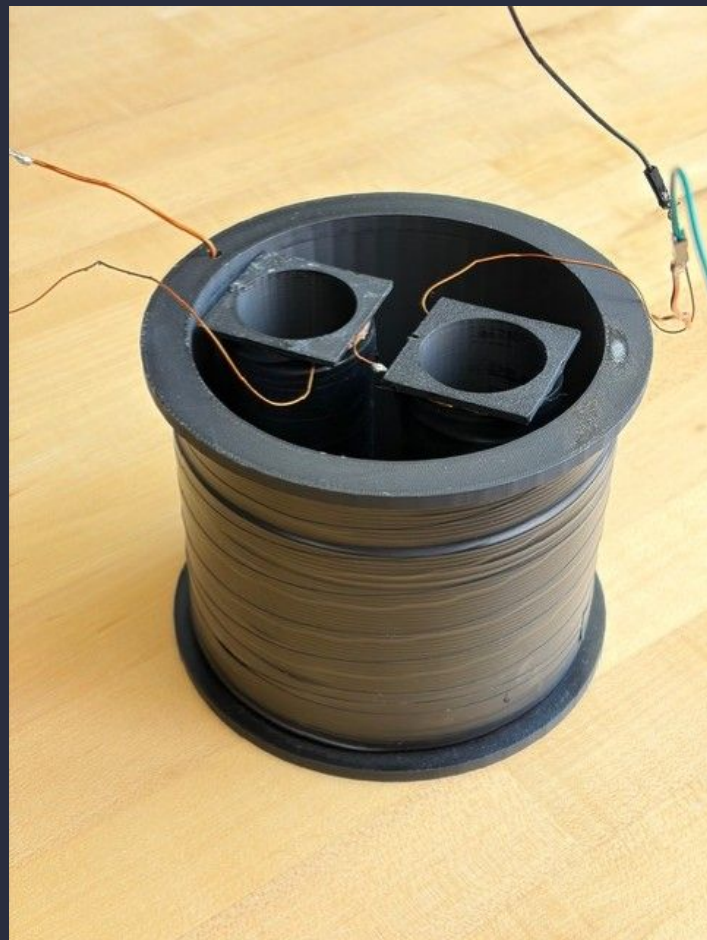
- 228 turns of 14AWG wire
- 4.6mH and 1.1 Ω
- 11.4cm I.D. and 10cm height

Calculated Performance:

- 12V pulse produces 10.9A current
- Produces ~30mT at the coil center

Noise Reduction Strategy:

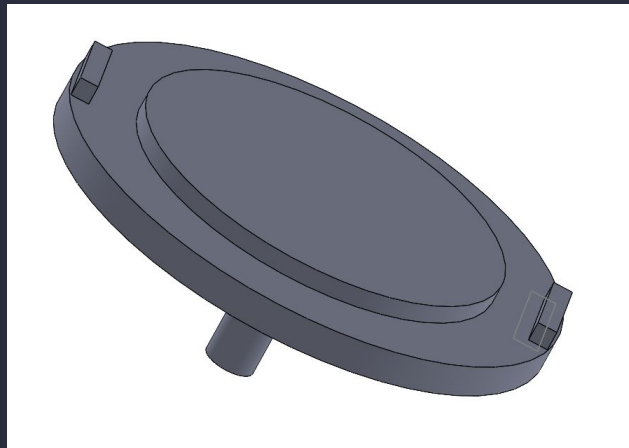
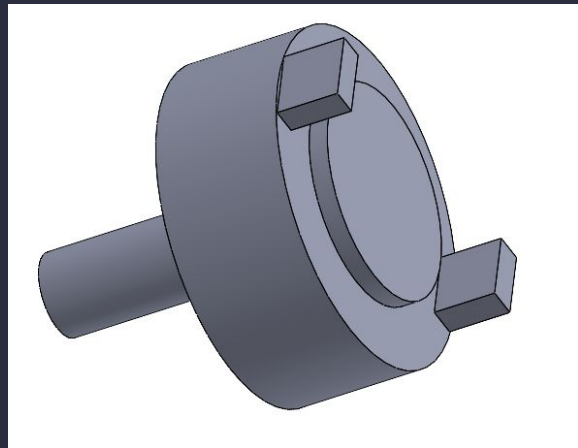
- Two identical pickup coils, one is a sample coil and the other a dummy coil
- Each coil has 552 turns of 22AWG wire
- Connected in a series-opposed circuit



Coil Winding Chucks

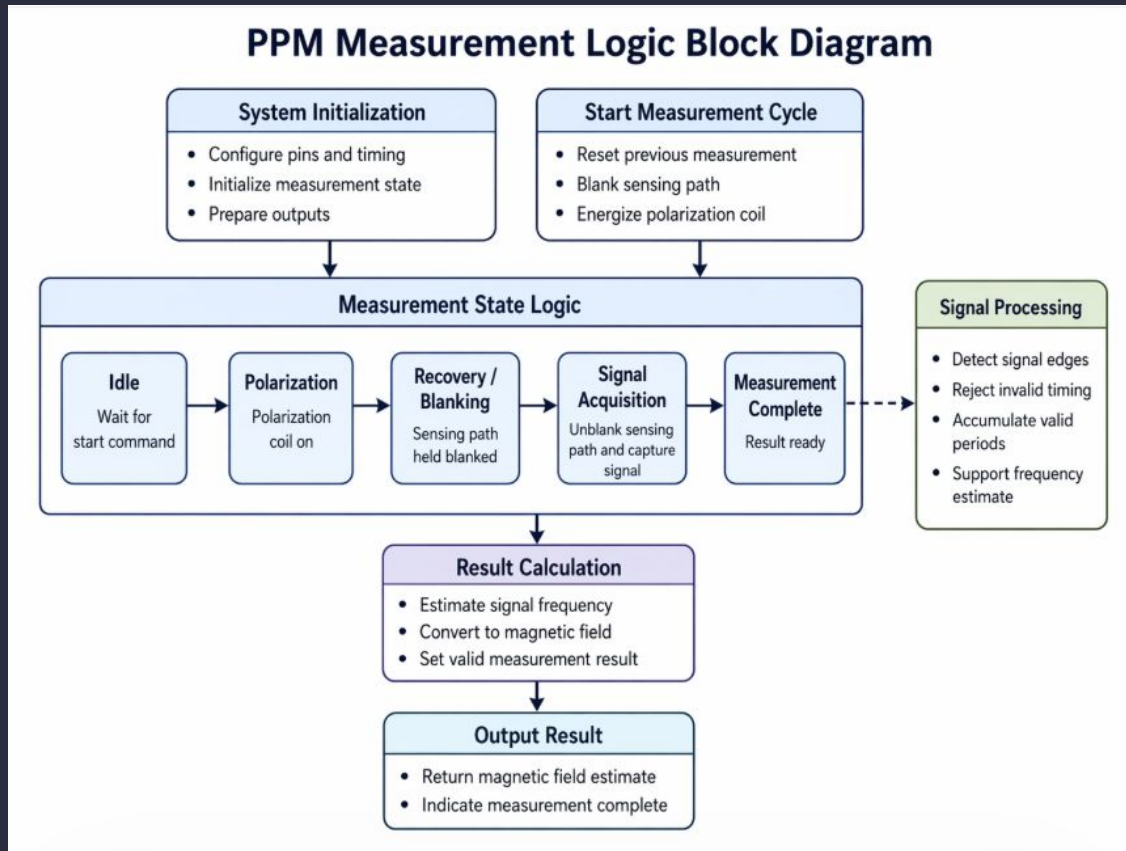
Polarizing Coil Chuck: Designed to hold the polarization coil in the coil winder.

Sensor and Dummy Coil Chuck: Designed to hold the sensor and dummy coils in the coil winder.



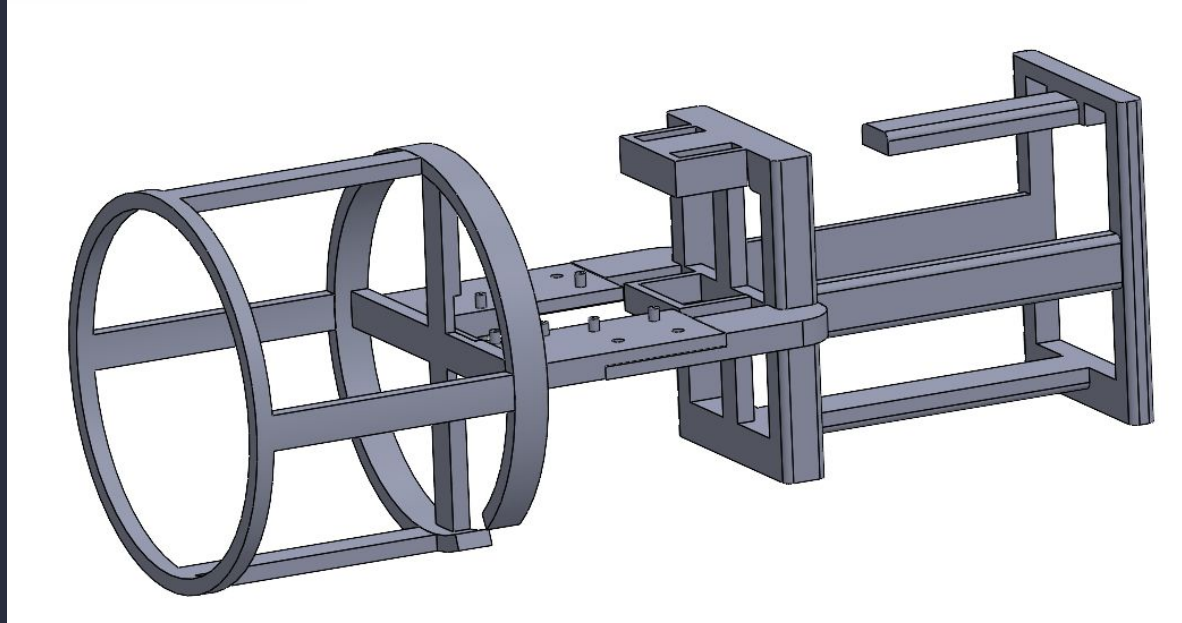
Magnetic Measurement Algorithm

The Teensy polarizes the sample, blanks and then waits for recovery, then times comparator pulses to estimate precession frequency and magnetic field. It compares the result to a baseline and flags anomalies over 150 nT.



Towfish Skeleton Model

Acts as an internal frame that fits snugly within a 6" PVC pipe to securely hold a 12V 5A lead acid battery, a 3s LiPo battery, the polarization and sensor coils, as well as the PCBs

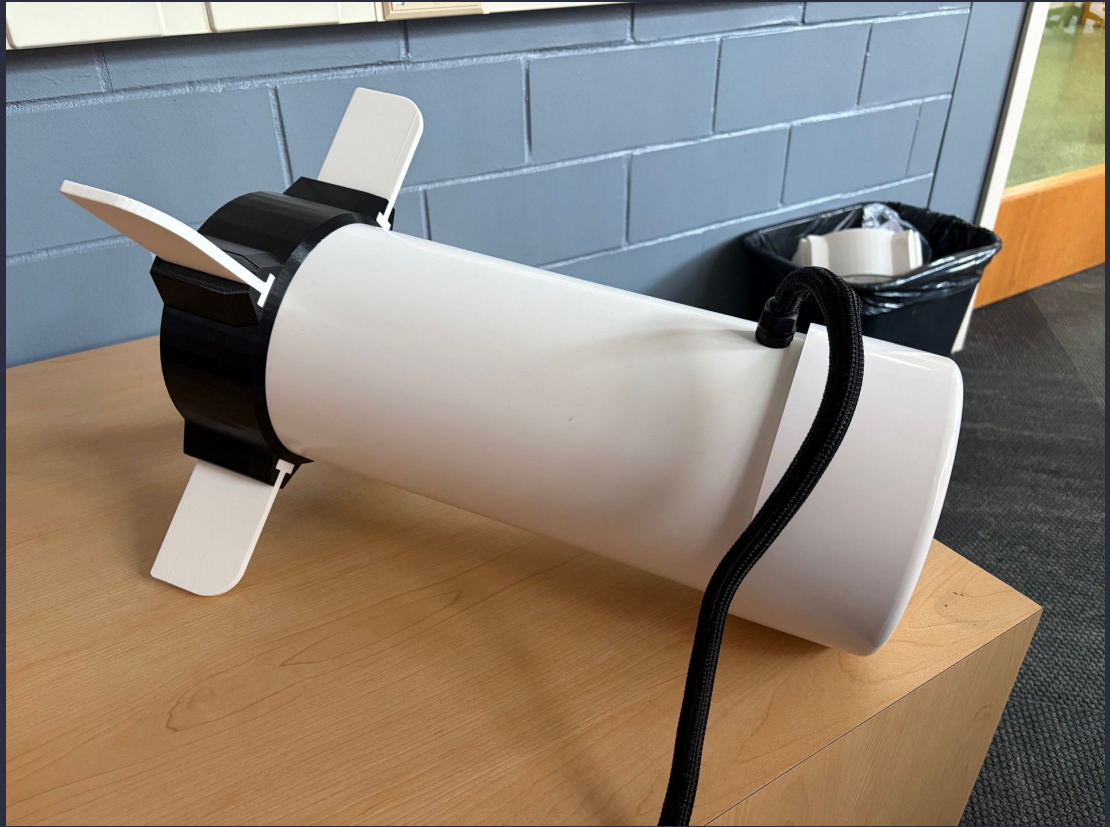


Towfish Design

Protects the magnetometer from impacts and water damage, as well as keeping the magnetometer and power sources afloat.

Made of 6" PVC fitted with a cap on the front. The rear has a 3D printed cap with built in fin attachment guards. It is attached by gorilla glue, and sealed with Flex Seal. Fins are also 3D printed and detachable for easy transportation and replacement.

The towfish is attached to the boat by a nylon rope encased in a cable sleeve. The sleeve is able to accommodate a cable to transmit data from the boat to the towfish if necessary.



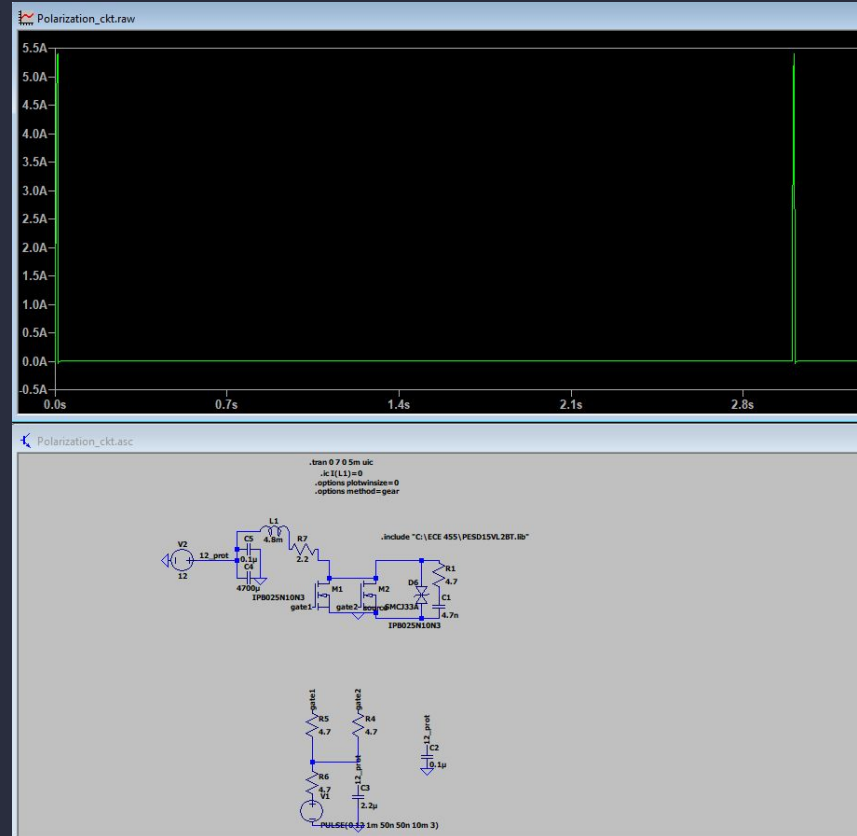
Results – Power Board

12V input and properly
produced 3.3V, +-5V,
and 12V protected rail.



Results – Pulse Controller

Pulse Controller - Pulsed 4.7A consistently while testing with 12V source with a 10ms pulse and a 1.2Ohm resistor in series the current was draining in approximately 5ms with minimal ringing.



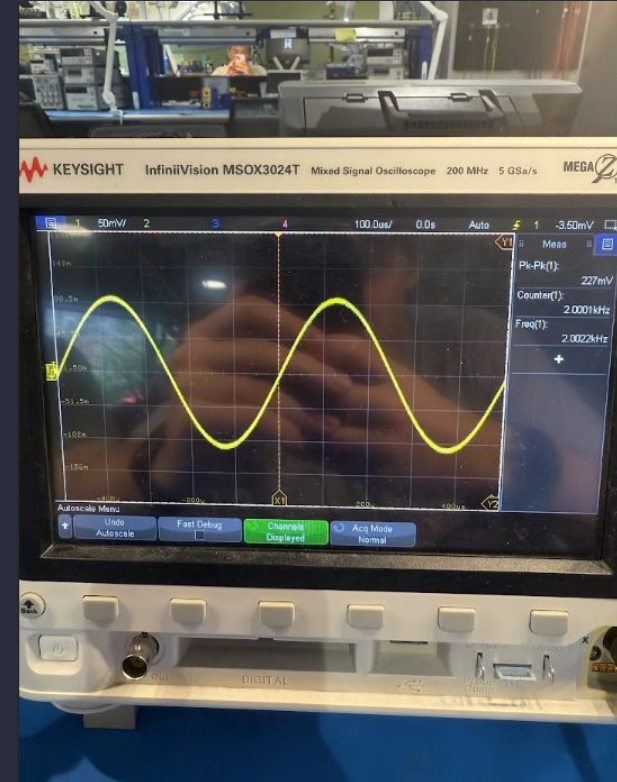
Results – Blanking & Pre Amplifier

Key Takeaway

- Blanking successfully protected the analog chain during switching transients
- Pre-amplifier produced expected $\sim 20\times$ gain

Test Conditions

- Wave generator input: 2 kHz, 10 mVpp



Results – Filtering

Test Conditions

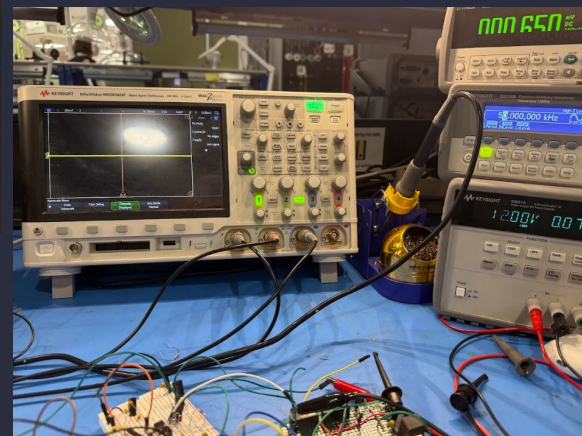
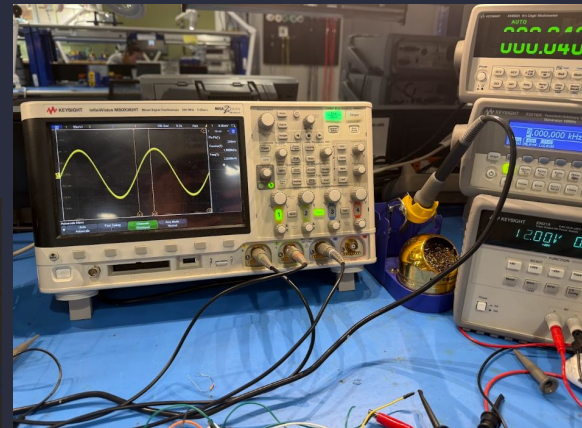
- Bandpass target: 800 Hz–4 kHz
- Swept 50 mVpp sine wave from 60 Hz to 60 kHz
- Checked attenuation below and above passband

Key Finding

- Passband signal remained visible
- Out-of-band signals were attenuated on both sides
- Filter supports expected proton signal range



Center image: Out-of-band attenuation check
Top right: Signal passed within filter band
Bottom right: Out-of-band attenuation check



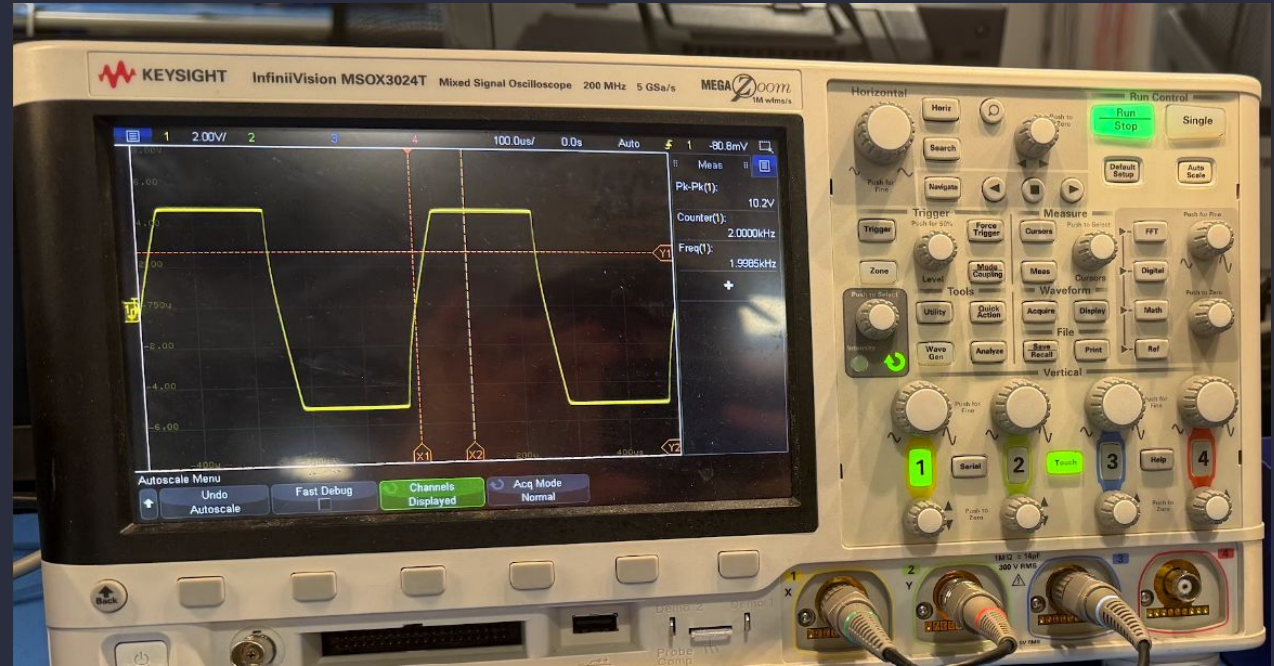
Results – Post Amplifier

Test Conditions

- Input: 2 kHz, 10 mVpp
- Post-amp gain: 100×
- Total chain gain: ~2000×

Result

- Output reached ~10 Vpp
- Clipping occurred at op-amp limits
- Confirms Amplifier is operational



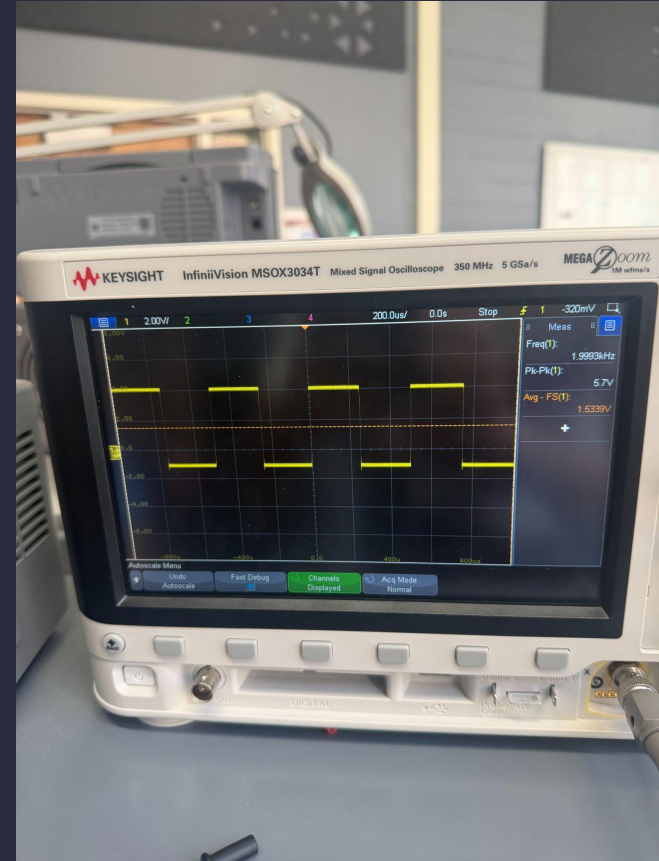
Results – Comparator & MCU

Test Conditions

- Input: 2 kHz
- Amp 10 mVpp
- Offset: 0mV

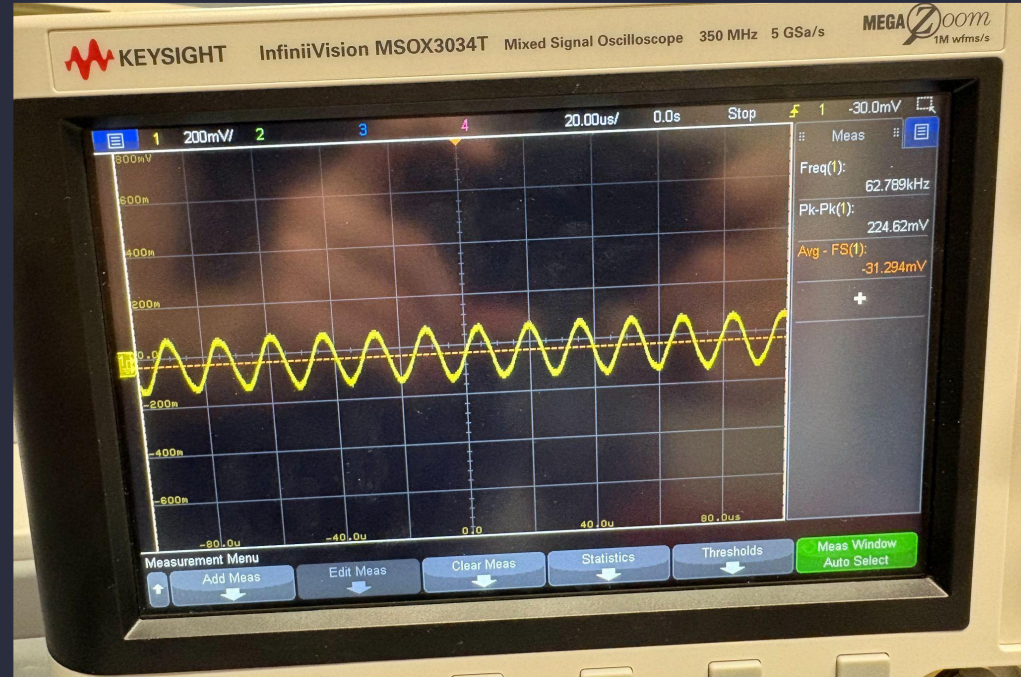
Result

- Generated a clean square wave 10mVp, 2 khz signal.
- Measured an output frequency which is about the same as the input.
- Stable operation across 800 Hz - 4 KHz input range.
- Output logic levels compatible with MCU input.



Results – Sensing Coil Test

- To verify sensing-coil operation, a second coil was driven with an AC signal and placed near the sensing coil.
- The sensing coil successfully detected the magnetic field generated by the driven coil, producing the waveform shown.
- Measured response: 62.789 kHz and 224.62 mV peak-to-peak, confirming that the sensing coil and readout path can detect an external coil-generated signal.



Challenges

Weak or Missing Proton Signal

The main challenge was detecting the very weak proton signal after the polarization pulse. at many points, the system showed noise, ringing, or test signals, but not a clear proton response. The signal strength was much smaller than expected compared to the noise.

What We Did

- Added in a multi-stage analog chain with pre-amplification, filtering, and post-amplification
- Added blanking to protect the receiver during the pulse and early recovery
- Investigated gain distribution across stages to improve sensitivity.

What Helped

- Stage-by-stage testing confirmed each block functioned independently
- Blanking reduced transient overload into the analog chain
- Filtering helped isolate the expected frequency range

What Still Needs Work

- Improve overall SNR enough to clearly recover the proton signal from the actual coil
- Use final PCB implementation to reduce parasitics from bread board and external pickup
- Continue optimizing recovery time, gain, and coil configuration

Ring & Blanking Challenges

High-Frequency Ringing After the Pulse

- Large ringing appeared at the coil output during and after the polarization pulse due to coupling between the sensing and polarization coils.
- Ringing often masked the expected signal

What We Did

- Added analog blanking at the start of the receive chain using the DG419 switch
- Adjusted blanking timing to occur before and after the pulse
- Tried extending recovery time before observing the receive path
- Tested damping approaches and monitored coil behavior directly on the oscilloscope

What Helped

- Blanking protected the analog front end from large switching transients
- Longer blanking windows reduced the amount of ring entering the amplification stages

What Still Needs Work

- Further optimize blanking timing relative to pulse shutoff
- Reduce ringing directly at the source through coil refinement

Noise & Interference Challenges

Challenge: 60 Hz and 60 kHz Environmental Noise

- Significant 60 Hz pickup was present when the system was not blanked
- Additional high-frequency interference near 60 kHz was also observed
- This was noise from the environment and lab equipment

What We Did

- Added a 4th order active band pass filtering to suppress noise
- Investigated dummy coil/common-mode cancellation strategies
- Physically moved coils and sensitive circuitry apart to reduce coupling

What Helped

- Active high-pass and low-pass filtering reduced low- and high-frequency interference
- Increasing physical separation reduced unwanted coupling into the receiver chain

What Still Needs Work

- Improve shielding and cable routing
- Further reduce environmental pickup through PCB layout and enclosure design
- Optimize dummy coil implementation for better common cancellation

Challenge: Converting the small proton precession signal into a digital wave (square wave) as input to the MCU to derive frequency.

- Initial circuitry unintentionally converted the sinusoidal FID signal into a DC-like value
- MCU requires frequency (zero-crossing timing), not amplitude or DC level
- Result: no usable magnetic field measurement

What We Did

- Redesigned front-end to preserve AC signal instead of rectifying it
- Implemented comparator with reference voltage (~1.65V) to detect zero crossings
- Added pull-up resistor to ensure valid digital HIGH output
- Added hysteresis (feedback resistor) to eliminate noise-induced switching
- Tested with function generator (800 Hz – 4 kHz, low mV signals)

What Helped

- Hysteresis stabilized comparator output and eliminated edge jitter
- Proper biasing (mid-rail reference) enabled clean zero-crossing detection

What Still Needs Work

- Further reduce parasitic coupling and ringing by testing the circuitry using our PCBs made.

Note:

Looking at previous challenges on the analog side. We were unable to detect or see a signal when we connected the coils into the circuitry due to heavy noise and ringing.

Since we couldn't see a decaying sinusoid after the coils, and into the amplifier and filtering circuitry. We could not test this board or convert the signal into the square wave for the MCU.

Looking at the slide on our results (slide 23), we used the wave gen which was a clean sine wave of frequency 2kHz, and it worked then.

Test the comparator and digital circuitry. Once a clean, decaying sinusoidal proton precession signal is produced from the coils, and is amplified and filtered through the analog and filtering stages.

Coil & Physical Layout Challenges

Challenge: Coil Behavior and Coupling to the Electronics

- Coil outputs showed ringing and sensitivity to physical setup
- Keeping the coil too close to the analog circuitry caused oscillation and interference
- Breadboard/perfboard implementation introduced extra parasitics and noise pickup

What We Did

- Extended the cable between the coils and analog circuitry
- Tested different loading/damping conditions on the coil
- Compared sensing-only and sensing + dummy coil configurations
- Began moving from loose prototyping toward dedicated PCB implementation

What Helped

- Separating the coil from the analog hardware reduced unwanted coupling
- Testing alternate configurations improved understanding of coil behavior – confirmed sensing + dummy coil configurations was reducing noise like intended.

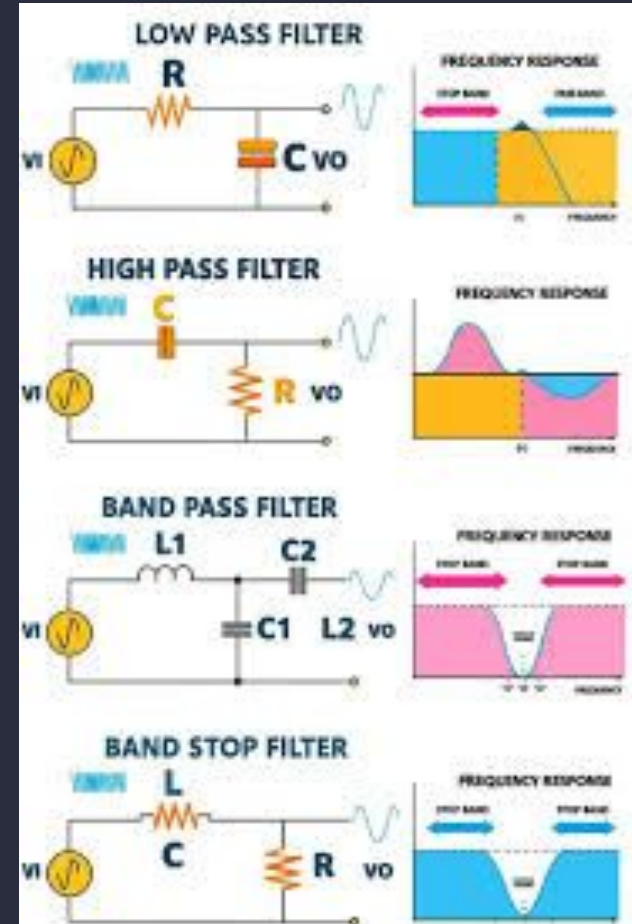
What Still Needs Work

- Finalize coil design and interface for best signal recovery
- Reduce parasitics and external pickup with the full PCB version

Future Considerations

Future Considerations

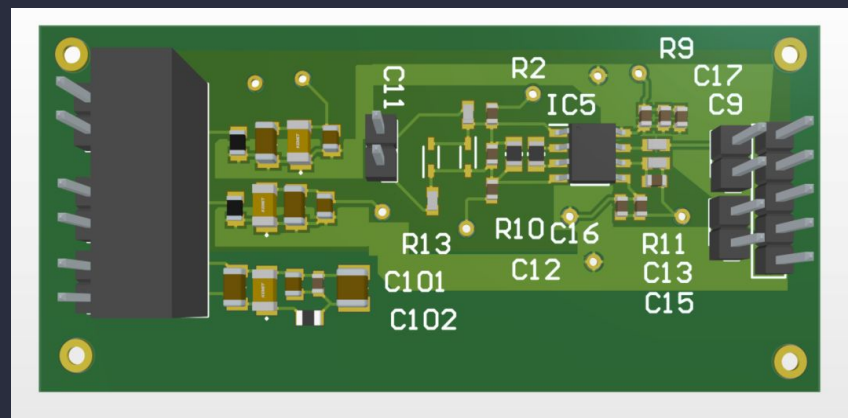
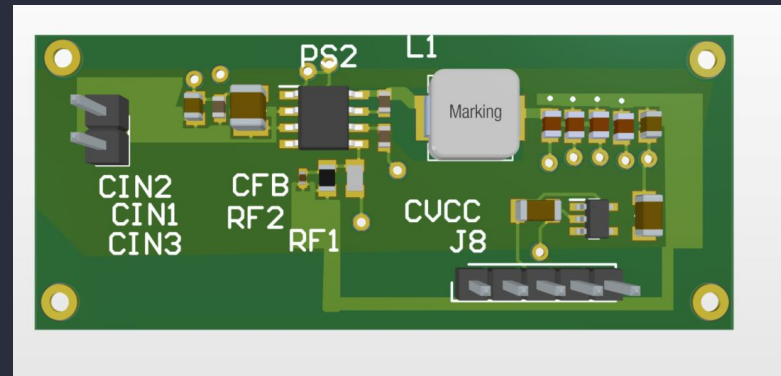
- Utilize perforated boards over breadboards for testing. To increase current capabilities and make solid connections.
- Test different filtering, gain strategies, and detection methods to improve sensitivity to weak proton precession signals.
- Redesign the coil geometry and placement.
- Elongate and enlarge towfish to provide more space for a coil redesign, avoid oscillation, and to promote increased buoyancy.



Transition from Breadboards to PCBs

May 2026

- Breadboards introduce loose connections, parasitic capacitance, and inconsistent contact resistance, which can distort weak analog signals.
- A PCB would provide more reliable current paths for the pulse circuitry and cleaner signal paths for the analog front end.
- Shorter traces and controlled wiring would reduce unwanted ringing, pickup, and coupling between the pulse and sensing circuits.
- More repeatable hardware would make troubleshooting easier because signal changes would be caused by the circuit, not unstable connections.



Improve Filtering, Gain Strategy, and Signal Detection

- Further testing should evaluate the best gain distribution between the instrumentation amplifier and active filter stages.
- Too much early gain can amplify noise before the desired FID signal is isolated.



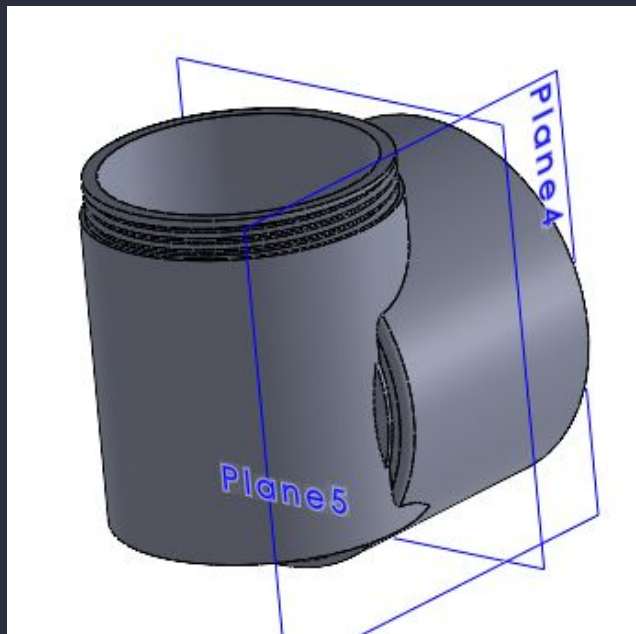
Enlarge Towfish Housing for Improved Layout

- A larger or elongated towfish would provide more room to physically separate the pulse circuitry, sensing coils, analog front end, and digital electronics.
- Increased spacing would reduce electromagnetic coupling between the high-current pulse circuit and the analog circuit.
- More internal volume would allow cleaner wire routing and reduce accidental loop areas that pick up noise.
- A redesigned housing could also improve buoyancy control, cable routing, and waterproofing.



Reoriented coil facing downward to more accurately detect sunk wreckage located below the towfish. Designed to be fitted on the end of the 6" PVC pipe as a temporary solution. Included is a screw on cap to secure the polarization coil in the holder.

Useful for if enlarging the towfish housing up from 6" PVC pipe is not feasible.



Add Stronger Pulse Isolation and Protection

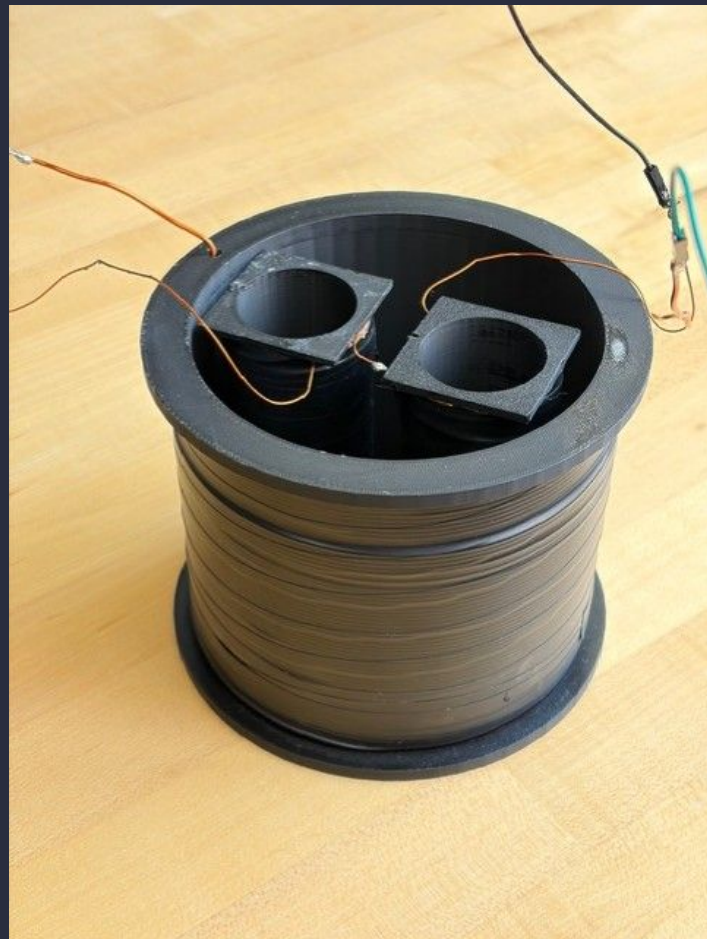
- The largest interference occurred during and immediately after the high-current polarization pulse.
- Future designs should add stronger isolation between the pulse coil and analog circuitry.
- In addition to blanking could add additional clamping, series resistance, or transient protection could help prevent large pulse induced voltages from reaching the amplifier.
- Improving pulse isolation could allow the receiver to recover faster and increase the chance of detecting the weak FID signal after the pulse.



Improved Differential Coil Matching

May 2026

- Refine sensing and dummy coil configuration for better noise cancellation
- Match coil turns, size, placement, and orientation between both coils
- Use the dummy coil to subtract shared 60 Hz and lab interference
- Should help isolate the proton signal



Bill of Materials

Towfish Design Bill of Materials

Cable pass through(Qty=1): (\$6.99) https://www.amazon.com/dp/B0GDTMZ4R?psc=1&ref_=cm_sw_r_cp_ud_ct_RJHM0184KV33Z3I1D4PQ

Nylon Rope (Qty=1): (\$6.95) https://www.amazon.com/dp/B007PBKN4K?psc=1&ref_=cm_sw_r_cp_ud_ct_RJHM0184KV33Z3I1D4PQ_1

SD Card (Qty=1): (\$39.99) https://www.amazon.com/dp/B07FCMKK5X?psc=1&ref_=cm_sw_r_cp_ud_ct_RJHM0184KV33Z3I1D4PQ_2

Cable Sleeve (Qty=1): (\$14.99) https://www.amazon.com/dp/B07FXFI2HC?psc=1&ref_=cm_sw_r_cp_ud_ct_RJHM0184KV33Z3I1D4PQ_3

PVC Cement (Qty=1): (\$11.97) https://www.amazon.com/dp/B0002YU23O?psc=1&ref_=cm_sw_r_cp_ud_ct_RJHM0184KV33Z3I1D4PQ_4

6" x 17"L PVC Pipe (Qty=1): (\$26.99) https://www.amazon.com/dp/B0FWRHPD8K?psc=1&ref_=cm_sw_r_cp_ud_ct_RJHM0184KV33Z3I1D4PQ_5

6" PVC End Cap (Qty=2): (\$14.99/ea) https://www.amazon.com/dp/B0BRKWK28F?psc=1&ref_=cm_sw_r_cp_ud_ct_RJHM0184KV33Z3I1D4PQ_6

M4 Screw Kit (Qty=1): (\$8.99) https://www.amazon.com/dp/B08FR6T6JL?psc=1&ref_=cm_sw_r_cp_ud_ct_RJHM0184KV33Z3I1D4PQ_7

M1 Screw Kit (Qty=1): (\$5.47) https://www.amazon.com/dp/B0GHO922OD?psc=1&ref_=cm_sw_r_cp_ud_ct_RJHM0184KV33Z3I1D4PQ_8

Magnetometer Circuitry Bill of Materials

[SP26_FindingTitan_BOM_Michael_Krueger.BomDoc](#)

[Proton_Magnetometer_Proto_BOM.xlsx](#)

[mrkrueger3/ECE455-Finding-Titan: ECE455 Finding Titan Documentation](https://github.com/mrkrueger3/ECE455-Finding-Titan)

Thank you